Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/06/2019
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Individual Differences in the Neuronal and Behavioral Response to Exposure to HZE Particles		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	08/01/2016	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 5/4/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Poulouse, Shibu Ph.D. (Tufts University School of Medicine) Shukitt-Hale, Barbara Ph.D. (US Department of Agriculture/Agricultural Research Service, Beltsville, MD)		
Grant/Contract No.:	NNX16AE06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to single, above threshold doses of HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neuronal and behavioral functioning after the conclusion of the mission. The effects of exposure to multiple, subthreshold doses of the same or different particles or protons have not been extensively studied. The objectives of the proposed research program are: (1) to evaluate the effects of repeated below threshold/above threshold exposures to the same and different HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative stress, inflammation, signaling molecules, and autophagy in key brain regions; and (3) to determine how individual characteristics, including age and sex, may influence the responsiveness of the individual to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our data indicate that there is a significant risk that exposure to low, NASA-relevant doses of HZE particles produces alterations in CNS function and in the behaviors that are dependent upon the integrity of specific CNS systems. However, current research suggests several caveats that might modulate the effectiveness of such exposures in disrupting neuronal function and cognitive performance. The experiments presented in this report were designed to evaluate several factors which may affect the degree to which exposure to space radiation might affect the cognitive ability of astronauts on exploratory class missions. The first two series of experiments utilize exposure to 4He particles because 4He and 1H particles will constitute the major source of radiation within the spacecraft during exploratory class missions. One series of experiments examines the effects of repeated exposure to a below threshold dose of 4He particles on a subsequent exposure to a second dose which is either below or above threshold. This experiment attempts to mimic the most likely pattern of radiation exposure to which astronauts will be exposed on exploratory class missions involves repeated exposures to below threshold doses. In this context, threshold is defined as the lowest dose of a specific particle which will produce a deficit in cognitive performance. These experiments presented in the current report using both subthreshold and above threshold doses of 4He particles suggest that the effects on cognitive performance may be task specific and may also vary as a function of the interval between exposures. Current experiments (6 month interval) combined with previous research (2, 24, or 72 hr intervals) suggest that a below threshold conditioning exposure can either enhance or prevent the effects if exposure to 4He particles depending upon the time interval and specific cognitive task. A second series of experiments evaluated the effects of exposure to 4He particles on the performance of female organisms and the possible use of antioxidant diets (blueberry) as neuroprotectants. Ovariectomized (OVX) female rats implanted with tubes containing estradiol or vehicle and intact rats maintained on control diets or antioxidant diets containing blueberry powder have been exposed to 4He particles. The present results using female subjects contrast with the results using male subjects: male rats maintained on blueberry diets show decreased responsiveness to the disruptive effects of HZE particle irradiation on cognitive performance; intact (non-OVX) female show subjects show an enhanced responsiveness to the disruptive effects of HZE particle irradiation on cognitive performance. These results suggest that the radiation protection requirements on exploratory class missions may differ for male and female astronauts. While additional research is clearly needed, future research on the use of neuroprotective diets must include both male and female subjects. The third series of experiments rats was designed to determine whether the disruption of neuronal function and cognitive performance following exposure to 56Fe particles resulted from direct effects of the particles on neurons (“hits”) or whether the neuronal and cognitive effects were due to indirect effects of oxidative stress and neuroinflammation, which can be produced by the effects of HZE particles throughout the entire body. If oxidative stress and neuroinflammation are the critical mediators there should be no differences in the behavioral or neuronal endpoints as a function of the location of the exposure (head-only or body-only). If direct effects on neurons mediate the differences in the effects of higher or lower linear energy transfer (LET) particles, we should see threshold differences as a function of particle LET with head-only, but not with body-only exposures. The data indicates that irradiation-induced changes in neuronal function and the development of cognitive deficits are independent of location: head-only, body-only, or whole-body exposures produce similar changes in neuronal function and cognitive deficits.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Peer-reviewed Journals	Rabin BM, Polouse S, Bielinski DF, Shukitt-Hale B. "Effects of head-only or whole-body exposure to very low doses of 4He (1000 MeV/n) particles on neuronal function and cognitive performance." Life Sci Space Res (Amst). 2019 Feb;20:85-92. https://doi.org/10.1016/j.lssr.2019.02.001 ; PubMed PMID: 30797437 , Feb-2019